

一、精确问题陈述

目标是严格说明 printed page 1104 中 equation (5.5) 是否能由此前的 estimates 推出。原问题要求：

证明或否定 claimed limit (5.5)，并且把 change of variables、cutoff choice、normalization constant、one-sided-limit argument、limiting estimate 全部写清楚。若原推导不成立，需要指出 exact gap，并给出 corrected claim。

设 backward heat kernel 为

$$\rho_{X_0, T}(F, t) = [4\pi(T-t)]^{-1} \exp\left(-\frac{|F-X_0|^2}{4(T-t)}\right).$$

Section 3 的 localized quantity 是

$$\Psi_\chi(t) = \int_{\Sigma_t} (\cos \alpha)^{-p} \chi(F) \rho_{X_0, T}(F, t) d\mu_t,$$

其中 $\chi \in C_0^\infty(B_{2r}(X_0))$ ，且 $\chi = 1$ on $B_r(X_0)$ 。令

$$Q_\chi(t) = e^{c_1\sqrt{T-t}} \Psi_\chi(t).$$

Theorem 3.3 给出 one-sided differential inequality

$$Q'_\chi(t) \leq -N_\chi(t) + c_2 e^{c_1\sqrt{T-t}},$$

其中 $N_\chi(t) \geq 0$ 是 nonnegative square terms 的 integral。

Section 5 的 parabolic rescaling 是

$$F_\lambda(x, s) = \lambda(F(x, T + \lambda^{-2}s) - X_0), \quad d\mu_s^\lambda = \lambda^2 d\mu_{T+\lambda^{-2}s},$$

并且

$$w_\lambda = (\cos \alpha_\lambda)^{-p}.$$

问题中的 printed expression 使用固定 scaled cutoff $\phi_R(F_\lambda)$ ，并声称对 fixed $-1 < s_1 < s_2 < 0$ ，

$$G_\lambda(s_2) - G_\lambda(s_1) \rightarrow 0.$$

二、主要数学进展

Proven : fixed-cutoff terminal limit 存在。

固定一个 original-coordinate cutoff χ 。由 $N_\chi(t) \geq 0$,

$$Q'_\chi(t) \leq c_2 e^{c_1 \sqrt{T-t}}.$$

右端在 terminal interval 上 integrable。定义 tail error

$$A(t) = \int_t^T c_2 e^{c_1 \sqrt{T-\tau}} d\tau.$$

则 $A(t) \rightarrow 0$ as $t \uparrow T$, 并且

$$(Q_\chi + A)'(t) = Q'_\chi(t) - c_2 e^{c_1 \sqrt{T-t}} \leq 0.$$

所以 $Q_\chi + A$ monotone nonincreasing。又因为 $Q_\chi \geq 0$ 且 $A \geq 0$, 它有 finite one-sided limit。于是

$$\lim_{t \uparrow T} Q_\chi(t) = L_\chi$$

存在且有限。

这说明 one-sided inequality 和 positive error term 都不是障碍 : positive error 由 tail integral 吸收。

Proven : corrected fixed-cutoff rescaling identity.

令

$$t_\lambda(s) = T + \lambda^{-2}s, \quad s < 0.$$

则

$$T - t_\lambda(s) = \lambda^{-2}(-s), \quad F - X_0 = \lambda^{-1}F_\lambda.$$

Gaussian exponent 满足

$$\frac{|F - X_0|^2}{4(T - t_\lambda(s))} = \frac{|F_\lambda|^2}{4(-s)}.$$

二维 area scaling 给出

$$[4\pi(T - t_\lambda(s))]^{-1} d\mu_{t_\lambda(s)} = [4\pi(-s)]^{-1} d\mu_s^\lambda.$$

Kähler angle 在 positive homothetic rescaling 下 invariant, 因此

$$(\cos \alpha_\lambda)^{-p} = (\cos \alpha)^{-p}.$$

所以 fixed original cutoff $\chi(F)$ 在 scaled variables 中变成

$$\chi(X_0 + \lambda^{-1}F_\lambda).$$

定义 corrected endpoint quantity

$$G_{\lambda}^{\chi}(s) = e^{c_1 \sqrt{T-t_{\lambda}(s)}} \int_{\Sigma_s^{\lambda}} (\cos \alpha_{\lambda})^{-p} \chi(X_0 + \lambda^{-1} F_{\lambda})(-s)^{-1} e^{-|F_{\lambda}|^2/[4(-s)]} d\mu_s^{\lambda}.$$

则 exact identity 是

$$G_{\lambda}^{\chi}(s) = 4\pi Q_{\chi}(t_{\lambda}(s)).$$

若使用 normalized factor $[4\pi(-s)]^{-1}$, 则等式变为

$$G_{\lambda}^{\chi}(s) = Q_{\chi}(t_{\lambda}(s)).$$

因此 4π 只是 common normalization factor, 不影响 zero-difference limit.

Proven : corrected two-time limit.

因为 fixed $s_i < 0$ 时

$$t_{\lambda}(s_i) = T + \lambda^{-2} s_i \uparrow T,$$

所以

$$Q_{\chi}(t_{\lambda}(s_1)) \rightarrow L_{\chi}, \quad Q_{\chi}(t_{\lambda}(s_2)) \rightarrow L_{\chi}.$$

于是

$$G_{\lambda}^{\chi}(s_2) - G_{\lambda}^{\chi}(s_1) = 4\pi(Q_{\chi}(t_{\lambda}(s_2)) - Q_{\chi}(t_{\lambda}(s_1))) \rightarrow 0.$$

epsilon proof 也直接成立 : 给定 $\varepsilon > 0$, 取 $\delta > 0$ 使得

$$|Q_{\chi}(t) - L_{\chi}| < \frac{\varepsilon}{8\pi} \quad \text{whenever } T - \delta < t < T.$$

取 λ 足够大, 使 $T - t_{\lambda}(s_i) < \delta$ for $i = 1, 2$. 则

$$|G_{\lambda}^{\chi}(s_2) - G_{\lambda}^{\chi}(s_1)| \leq 4\pi|Q_{\chi}(t_{\lambda}(s_2)) - L_{\chi}| + 4\pi|Q_{\chi}(t_{\lambda}(s_1)) - L_{\chi}| < \varepsilon.$$

三、主要障碍

真正的 gap 是 cutoff.

fixed original cutoff $\chi(F)$ 在 scaled variables 中对应

$$\chi(X_0 + \lambda^{-1} F_{\lambda}),$$

而 printed equation 使用的是 fixed scaled cutoff

$$\phi_R(F_{\lambda}).$$

后者在 original variables 中是

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)),$$

其 support 在 $B_{2R/\lambda}(X_0)$, 随 λ 改变。

因此 printed expression 不是同一个 fixed quantity Q_χ 的两个 terminal evaluations , 而是一族 moving-cutoff quantities。对每个 λ 单独有 terminal limit , 并不能推出

$$Q_\lambda(T + \lambda^{-2}s_2) - Q_\lambda(T + \lambda^{-2}s_1) \rightarrow 0.$$

还需要 uniform moving-cutoff estimate 或 cutoff-error theorem。此前 estimates 没有提供这一点。

四、方法时间线

stage	question addressed	conclusion established	effect on the approach
1	Does the one-sided differential inequality imply a terminal limit?	Yes. Adding the integrable tail error makes the quantity monotone and bounded.	The positive error term is harmless.
2	What is the exact parabolic scaling identity?	The unnormalized scaled Gaussian integral equals 4π times the normalized Section 3 quantity.	The 4π factor is bookkeeping only.
3	What cutoff is produced by a fixed Section 3 cutoff?	It becomes $\chi(X_0 + \lambda^{-1}F_\lambda)$ in scaled variables.	This gives a rigorous corrected version of (5.5).
4	Does this equal the printed $\phi_R(F_\lambda)$?	No. $\phi_R(F_\lambda)$ is a moving original cutoff $\phi_R(\lambda(F - X_0))$.	The printed derivation has a cutoff gap.
5	Can the corrected statement still serve local estimates on $B_R(0)$?	Yes. If $\lambda r \geq R$, then $\chi(X_0 + \lambda^{-1}Y) = 1$ on $B_R(0)$.	Later local nonnegative integrals can still be controlled.

五、当前状态与下一步

问题按 printed form 目前未由前文 rigorously proved。严格可得的是 fixed-cutoff corrected statement :

$$G_\lambda^\chi(s_2) - G_\lambda^\chi(s_1) \rightarrow 0.$$

printed cutoff version 需要补充以下 remaining lemma :

Remaining lemma. Let $\phi_R \in C_0^\infty(B_{2R}(0))$ with $\phi_R = 1$ on $B_R(0)$, and define $\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0))$. For fixed $-1 < s_1 < s_2 < 0$, prove a uniform moving-cutoff convergence theorem showing that

$$e^{c_1 \sqrt{T-t_\lambda(s_2)}} \Psi_{\chi_{R,\lambda}}(t_\lambda(s_2)) - e^{c_1 \sqrt{T-t_\lambda(s_1)}} \Psi_{\chi_{R,\lambda}}(t_\lambda(s_1)) \rightarrow 0$$

as $\lambda \rightarrow \infty$, where $t_\lambda(s) = T + \lambda^{-2}s$. Equivalently, prove that the moving cutoff family has a uniform convergence-to-terminal-limit estimate strong enough at the parabolic scale $T - t \sim \lambda^{-2}$.

没有这个 lemma , (3.6) 只能推出 corrected fixed-cutoff version , 不能推出 printed fixed scaled-cutoff formula。